

BOMI KWON

Seoul, Republic of Korea · bomi9970@sookmyung.ac.kr · kwonbomi.github.io

Research Interests: Structured and controllable generative modeling, diffusion models, and text-conditioned generation.

EDUCATION	Sookmyung Women's University	Seoul, Republic of Korea
	<i>Master of Science, Department of Computer Science</i> GPA: 4.41/4.5	Mar. 2025 – Feb. 2027 (expected)
	<i>Bachelor of Science, Department of Statistics</i>	Mar. 2020 – Feb. 2025
	<i>Bachelor of Business Administration, Department of Business Administration</i> Total GPA: 3.61/4.5	
PUBLICATIONS	B. Kwon and K. Y. Lee (2026). Text-Conditioned Scene Graph Sequence Generation via Discrete Diffusion. <i>Proceedings of the 35th ACM International Conference on Information and Knowledge Management (CIKM '26)</i> . Accepted Short Paper.	
	B. Kwon and K. Y. Lee (2026). Text-driven 2D Human Pose Generation via Diffusion in Directional Angle Space. <i>The Transactions of the Korea Information Processing Society</i> , 15(7), 610–619.	
HONORS & AWARDS	Principal Investigator (Research Leader), 2026 STEM Research Team Project — General Track	
	<i>Korea Foundation for Women in Science, Engineering and Technology (WISET)</i> • Selected as one of 50 research teams nationwide ; awarded KRW 7,000,000 in competitive research funding.	Apr. 2026 – Oct. 2026
	Sookmyung Leadership Activity Scholarship	Jan. 2023
PRESENTATIONS	B. Kwon and K. Y. Lee (2026). Diffusion-based Skeleton Pose Editing. 2026 Spring Conference of the Korean Institute of Smart Media and the Korean Society of Digital Industry.	
	B. Kwon and K. Y. Lee (2025). A Graph Generation Model for Text-based Graph Editing. Korea Software Congress 2025 (KSC 2025).	
	B. Kwon and K. Y. Lee (2025). Development of a Large-Scale Graph Retrieval System based on LLM and Graph Database Providing a Natural Language Interface. Korea Computer Congress 2025 (KCC 2025).	
	B. Kwon, S. Park, Y. Park, and K. Y. Lee (2026). Text-Conditioned Diffusion for Static Humanoid Robot Pose Generation with Joint-Limit Embedding and Self-Collision Guidance. <i>Submitted to KICS 2026.</i>	
RESEARCH EXPERIENCE	Sookmyung Women's University	Seoul, Republic of Korea
	Graduate Researcher, Data Intelligence Laboratory	Mar. 2025 – Present
	• Conduct research on generative AI, with a focus on diffusion models, structured generation, and text-conditioned generation. • Design data pipelines, model architectures, experimental protocols, and evaluation frameworks for generative modeling research. • Implement baselines, analyze experimental results, and prepare manuscripts for journal and conference submissions.	
	Korea Foundation for Women in Science, Engineering and Technology (WISET) Principal Investigator (Research Leader), STEM Research Team Project	
	Apr. 2026 – Oct. 2026	
	• Lead a research team of two undergraduate students developing synthetic human-pose data generation methods for industrial safety research. • Manage research planning, experimental direction, budget execution, and coordination with participating members and related institutions. • Mentor undergraduate and secondary-school students in generative AI, diffusion models, dataset construction, and research methodology.	

PROJECTS **Text-Conditioned Scene Graph Sequence Generation via Discrete Diffusion** 2025 – 2026

- Developed a discrete diffusion framework for generating time-ordered scene graph sequences from natural-language descriptions of human–object interactions.
- Designed a Transformer-based model to capture object relationships within each frame, temporal changes across frames, and alignment with the input text; constructed training data from **Charades-STA** and **Action Genome** and evaluated the framework against multiple retrieval and generative baselines.
- **Led the project end-to-end as first author**, including problem formulation, model design and implementation, experimental design and analysis, and manuscript preparation.

Text-Conditioned Humanoid Robot Pose Generation with Physical Constraints 2026

- Developed a text-conditioned diffusion model for generating physically valid static poses for the **Unitree G1** humanoid robot.
- Retargeted human motion data to Unitree G1 in **MuJoCo** and incorporated joint-range and self-collision constraints into the generation process.
- Designed constraint-aware generation methods to improve physical validity while maintaining text-based pose control and diversity.

Text-driven 2D Human Pose Generation via Diffusion in Directional Angle Space 2025 – 2026

- Proposed a directional-angle representation for text-driven 2D human pose generation, representing each body segment by its directional angles rather than Cartesian joint coordinates.
- Built a text-conditioned diffusion pipeline using **BGE-M3** representations to generate poses while preserving consistent skeletal proportions.

AISUM Industry–Academic Collaboration — Large-Scale Product Matching and Retrieval System 2025 – Present

Product Matching (PM)

- Contributed to an image-based product matching system combining **object detection and visual embedding models** for similar-product retrieval.
- Evaluated multiple image embedding models and ensemble strategies and developed object-aware ranking methods for product search.

Semantic Matching (SM)

- Developed a semantic retrieval system for recommending products that are contextually relevant to the content and intent of online articles.
- Used **LLM prompt engineering**, **BGE-m3-ko/KURE-v1 embeddings**, and **Qdrant** to build large-scale product representations and semantic search pipelines.

Agentic Retrieval System

- Integrated Product Matching and Semantic Matching functions into an **MCP-based tool interface** for use with LLM clients including GPT, Gemini, and Claude.
- Implemented a **LangGraph-based agent workflow** to orchestrate tool selection and multi-stage product retrieval, including automated follow-up searches.

Development of a Large-Scale Graph Retrieval System based on LLM and Graph Database Providing a Natural Language Interface 2025

- Developed a natural-language graph retrieval system that converts user queries into **Cypher queries** using an LLM and searches large-scale graph data stored in **Neo4j**.
- Built an end-to-end pipeline for graph generation, metadata-based storage, retrieval, and real-time visualization.
- Evaluated the system using natural-language queries with varying levels of complexity and ambiguity.

ENTREPRENEURIAL **fitapat (ZOOC)**
EXPERIENCE

Founding Team Member / Product & Marketing Lead

Jul. 2023 – Aug. 2024

- Originated the product concept from an idea developed through SOPT and helped build the founding team to launch **fitapat**, an AI-based pet customization service.
- Led product and app/web planning for an end-to-end service that used generative image AI to create personalized pet characters from user-provided photos and connect the generated content to customized merchandise production.
- Led marketing and brand operations through product launch; developed the product experience around an AI customization system capable of generating **80+ character concepts from**

eight pet photos, and the fitapat platform and its physical products were showcased at CES 2024.

LEADERSHIP & ACTIVITIES

SOPT (Shout Our Passion Together)

Nationwide University Student IT Venture & Startup Organization

Product Manager, 31st Cohort

Jul. 2022 – Jan. 2023

Vice President, 32nd Cohort

Jan. 2023 – Jul. 2023

- Served as Vice President of the 32nd Cohort and previously worked as a Product Manager in the 31st Cohort, where I planned **ZOOC**, a digital service for preserving memories between pets and their families.

Sookmyung Admissions Ambassador “Polaris,” 16th Cohort

Sookmyung Women’s University, Office of Admissions

Vice President

Mar. 2021 – Feb. 2023

- Led university admissions and outreach activities, including prospective-student mentoring and admissions programs.

Statistics Education Talent Donation Activity

Statistics Korea (KOSTAT) — University Student Statistics Education Talent Donation Team

Volunteer Educator

May 2021 – Dec. 2021

- Provided statistical education to elementary, middle, and high school students.

SOOK-TAT, Department of Statistics Data Analysis Society

Sookmyung Women’s University, Department of Statistics

Member

Sep. 2022 – Jul. 2023

- Participated in regular data analysis studies and team-based activities.

Department of Statistics Student Council

Sookmyung Women’s University, Department of Statistics

Member

Mar. 2020 – Dec. 2021

- Participated in regular meetings and organized departmental events and student activities.

KB Campus Star, 17th Cohort

KB Kookmin Bank

Student Ambassador

May 2022 – Nov. 2022

- Created promotional content and participated in outreach activities for university students.

Amorepacific × Sookmyung Women’s University A.I. Team Project

Amorepacific Foundation

Oct. 28, 2024 – Dec. 6, 2024

- Completed a team project using generative AI to create **cosmetic concept boards and advertising video content**.

TECHNICAL SKILLS

Programming Languages: Python, R, SQL

ML & Frameworks: PyTorch, CUDA, LangChain, LangGraph, pandas, NumPy

Generative AI & LLM Systems: Diffusion Models, Transformer-based Models, LLM Tool Calling, MCP

Databases & Vector Search: Qdrant, PostgreSQL/pgvector, Neo4j

Simulation: MuJoCo

Developer Tools: Git, GitHub, Visual Studio Code, Overleaf

Certifications: ADsP (Advanced Data Analytics Semi-Professional)